

# Impedance Reduction Control of a Knee Joint Human-Exoskeleton System

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**Abstract**—Reducing human lower limb effort during walking is still a challenge for wearable exoskeletons. From the wearer's viewpoint, joint impedance increases while wearing the exoskeleton due to the fact that the exoskeleton's mass, friction, and gravity are added to those of the human lower limbs. Active impedance is required from the exoskeleton to compensate not only the exoskeleton's mechanical impedance but also that of the human limb. This paper presents a nonlinear impedance reduction control (IRC) approach applied to a knee joint exoskeleton during flexion/extension movements. The IRC approach ensures impedance adaptation of the human-exoskeleton system toward a desired level. The proposed method is an alternative to sensor-based impedance controllers where the human joint torque is estimated using electromyography (EMG) sensors, force/torque sensors, and so on. The performance of the proposed approach and its robustness with respect to modeling uncertainties are theoretically analyzed and discussed. Experiments were conducted with four healthy subjects to verify the effectiveness of human effort reduction. The root mean square of the EMG signals of the muscles involved in the flexion/extension was used as a metric. The experimental results show that EMG signal levels can be efficiently reduced by inertia, damping, and gravity compensation compared with those obtained without wearing the exoskeleton.

**Index Terms**—Assistive robotics, exoskeletons, human-exoskeleton interaction, impedance reduction control (IRC).

## I. INTRODUCTION

IN THE past few decades, an increasing number of exoskeleton devices have been developed to provide human performance augmentation either for daily living activities or for rehabilitation purposes [1]–[6]. In terms of powered assistive exoskeletons, one of the main goals is to reduce the wearer's effort during a movement initiated by the wearer without sacrificing his/her control priority, particularly for the wearer with partial movement capabilities, such as the elderly and patients affected by muscular weaknesses and/or movements disorders. However, interaction between the wearer and an assistive device is greatly limited by the inherent mechanical impedance of the coupled human-exoskeleton system.

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Important human efforts are required for achieving a desired movement if the exoskeleton interacting with the human limbs has a relatively high impedance.

One intuitive method to reduce the required human efforts to achieve a given task while wearing a powered exoskeleton is to reduce the impedance of the human-exoskeleton system through an appropriate control strategy [7], [8]. Another way is to compensate or to amplify the wearer's motor ability to achieve a desired movement. To a certain extent, the two above-mentioned methods are equivalent, since reducing the impedance of the human-exoskeleton system is beneficial in providing human power assistance. Woo and Kong [9] proposed an impedance reduction method of exoskeletons using a feedback controller to modulate the magnitude and phase of the impedance. In [10], a positive feedback approach was developed to increase the closed-loop exoskeleton system sensitivity to the wearer's forces and torques in order to reduce the wearer's effort for carrying a payload with wearing the Berkeley Lower Extremity Exoskeleton. However, using these two methods, the human effort cannot be effectively reduced compared with the effort without wearing exoskeleton and payload, since the apparent impedance of the exoskeletons are still passive. In other words, to guarantee a human effort reduction, an active impedance is required from the exoskeleton to compensate not only the exoskeleton's mechanical impedance but also the one resulting from the human limbs.

Ronsse *et al.* [11] proposed an oscillator-based method to provide assistance while wearing an exoskeleton and conducting cyclical movements. Adaptive oscillators are used to predict the desired human movement, while a virtual stiffness factor is used to generate the assistive torque. The results showed that the wearer's metabolic cost and the muscular activity can be both reduced during a given cyclical task, such as elbow cyclical movements or walking. In our previous study [12], an electromyography (EMG)-based method was proposed to amplify the wearer's joint torque during flexion/extension movements of the knee joint. Lee and Sankai [13] used the EMG-based method to adjust the impedance around knee joint with the objective of reducing the wearer's generated muscular activities. Aguirre-Ollinger *et al.* [14]–[16], proposed a torque-sensor-based method to compensate human lower limb impedances, such as damping and gravity, and a model-based method to compensate the inertia. In their study, the authors dealt with the agility of the wearer without correlating their approach with a reduction in the wearer's muscular activity.

In addition, it is also necessary to be able to independently modulate a single component of the joint impedance, such as the inertia, damping, and stiffness, during daily living activities, such as walking. A primary advantage is to provide different impedance assistances according to gait modes and phases. For example, [17] presents a control approach to facilitate recovery of walking, following stroke, by separately providing gravity compensation, joint stiffness, and damping reinforcement according to a gait phase state machine. It can also be noted that without appropriate compensation for inertia, the walking speed will be highly impacted, and the required human effort will increase consequently. Although many existing impedance reduction methods could effectively lower the whole impedance magnitude of the human-robot systems (see [9]) or augment the human muscular torque (see [18], [19]), few of them are able to increase/decrease each single component of the impedance (i.e., the inertia, damping, or stiffness) based on a generalized control structure. Although compensation of the gravity and friction of the coupled human-exoskeleton system is feasible, compensation for the system's inertia is quite challenging due to stability issues [20], [21].

In this paper, we propose an impedance reduction control (IRC) approach of a knee joint exoskeleton. The proposed approach is able to adapt the impedance model of the human-exoskeleton system to a desired level that is smaller than that of the original human limb impedance. Additionally, the proposed approach is able to compensate not only for gravity and damping of the human shank and foot but also for the inertia. The human joint torque is estimated using a nonlinear observer instead of EMG, force/torque sensors, and so on. In practice, the EMG-based approaches are usually very complex since the EMG signals from all associated muscles are generally needed, and the method is sensitive to many factors, such as the sensor positions, calibrations, and parameter identification [22]. In terms of use of torque/force sensors, the applicable places are constrained for many compact exoskeletons or orthoses. Additionally, in general, the torque/force sensors are subject to breakdowns and are expensive. Based on the estimated human joint torque, a desired knee joint movement is generated using a desired impedance model, while a nonlinear disturbance observer (NDO)-based sliding model controller is used to ensure satisfactory tracking of the desired knee joint human movements. Compared with the existing control methods, such as proportional-integral-derivative, linear quadratic control, and adaptive control, the NDO-based sliding-mode-based controller (SMC) has a significant advantage with respect to the position tracking accuracy and robustness against modeling errors and external disturbances (e.g., human torques) [23].

In the experiments, first, an experimental validation is conducted using a knee-joint-actuated exoskeleton that provides an assistance of the shank and foot in the sagittal plane to perform flexion/extension movements in sitting position. Different impedance states, such as human joint impedance state, human-exoskeleton impedance state, exoskeleton impedance compensation state, and human impedance compensation state, are designed. The root mean square (RMS) of the EMG signal is used as a metric to reflect the changes in the

human effort when performing similar joint movements with different impedance states. The experimental results show that the original inertia, damping, and gravity of the human joint can be effectively reduced. Furthermore, the proposed method is used to assist in swing motion during walking. The experimental results show that the human effort can also be reduced compared with the state without wearing the exoskeleton. To the best of our knowledge, the proposed approach, in which a nonlinear general torque-sensor-less control method is proposed to reduce the human muscular activity using inertia, damping, and stiffness compensation, is unique in the field of robot-aided physical assistance.

This paper was presented in part at the 2015 IEEE International Conference on Rehabilitation Robotics [24], where the active impedance control scheme was proposed for assisting the wearer in knee joint flexion/extension movements, and the preliminary experimental results were presented and showed that the original impedance of the human-exoskeleton system can be adapted to a relatively lower level. In this paper, we further investigate the impedance reduction of the human-exoskeleton system to ensure that the desired impedance is lower than the original impedance of the human limb. Additionally, a new human-exoskeleton model is proposed to extend the potential applications of the proposed method, and the new theoretical analyses, such as the performance analysis and the robustness with respect to modeling uncertainties, are added and discussed. Moreover, new detailed comparison experiments (e.g., different impedance states and generalization of the obtained results with different subject profiles) are implemented to verify the effectiveness of the proposed control strategy using an EMG-based metric. Furthermore, the proposed method is evaluated in a new daily living motion, i.e., the swing motion during walking.

The rest of this paper is organized as follows. Section II describes the human-exoskeleton system. In Section III, an analysis of the effects of wearing the exoskeleton on the impedance is presented. The detailed design of the proposed control structure and the associated theoretical analyses of its performance, stability, and robustness are given as well. In Section IV, the proposed impedance reduction method is verified by an experimental validation in knee-joint flexion/extension in sitting position with four healthy subjects. Moreover, an experimental validation of the proposed method in swing motion during walking is presented in Section V. Finally, a discussion and conclusions are presented in Sections VI and VII.

## II. HUMAN-EXOSKELETON SYSTEM

### A. Description of the Exoskeleton System

A single degree-of-freedom prototype exoskeleton [Exoskeleton Intelligently COMMunicating and Sensitive to Intention (EICOSI)] is designed to provide power assistance at the wearer's knee joint level (see Fig. 1). The exoskeleton consists of two segments attached separately to the thigh and shank and fixed to the wearer's lower limb using appropriate braces. It is driven by a high-power brushless dc motor (Maxon, Switzerland). To obtain a compact and

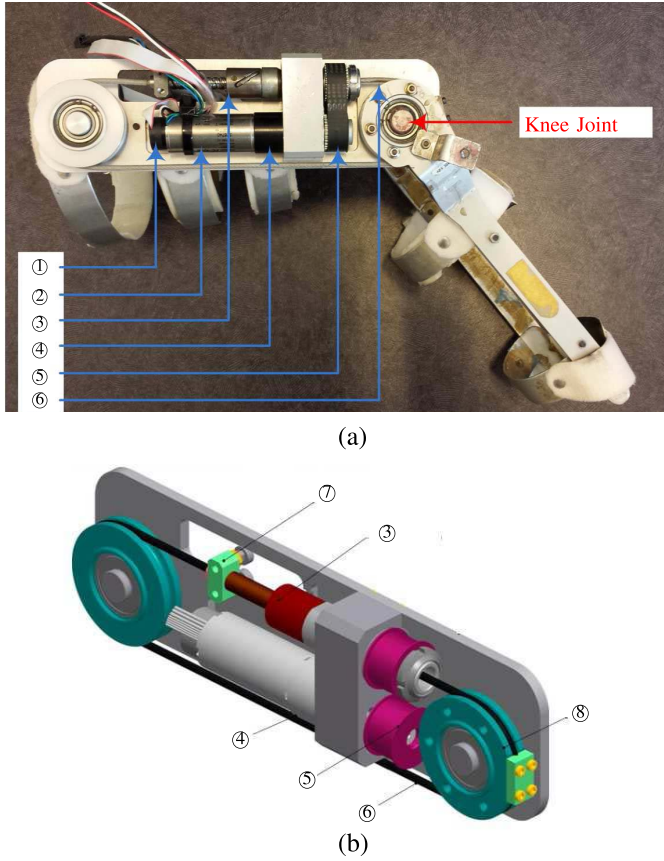


Fig. 1. EICOSI (a) knee joint EICOSI exoskeleton and (b) mechanical diagram of the actuation part. ①—encoder. ②—brushless dc motor. ③—ball screw. ④—gear motor. ⑤—belt transmission. ⑥—cable drive. ⑦—angle keeper. ⑧—pulley.

portable structure as well as a relatively high output torque, a compact transmission system is designed using a gear motor, a ball screw, a belt transmission, and a cable drive (see Fig. 1). The reduction ratio from the motor side to the joint side is 264:1. The whole actuator can deliver up to approximately 18 Nm. As a result of the exoskeleton armature, a certain friction is added to the knee joint.

The motor is equipped with an incremental encoder that measures the motor's rotation angle with a resolution of 1000 pulses per revolution. The knee joint angle can be calculated based on the reduction ratio of the whole transmission system. The angular velocity is obtained by a simple derivation of the joint angle. A fourth-order Butterworth filter is used to process the raw measured knee joint angle. The EICOSI is controlled using a host PC equipped with a controller board (dSPACE, Germany) running at 1 kHz. The control programs were developed using MATLAB (Mathworks, USA).

### B. Human-Exoskeleton Model

According to the Lagrange formulation, a generic dynamic model of the human lower limb coupled to a knee-joint exoskeleton during daily living activities can be written as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + \tau_f = \tau + J^T F \quad (1)$$

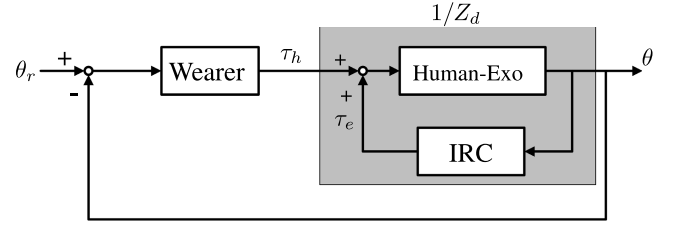


Fig. 2. Human-in-loop control structure with the exoskeleton's power assistance. IRC denotes the impedance reduction controller, and Human-Exo shows the human-exoskeleton system.  $Z_d$  represents a desired impedance.

where  $q$  represents the state vector of joint angles,  $M(q)$  is the inertia matrix,  $C(q, \dot{q})$  represents the Coriolis and centrifugal forces, and  $G(q)$  is the gravitational torque.  $\tau_f$  represents the friction torque, and  $F$  denotes the external force acting at the system, such as seat forces or ground reaction forces.  $J$  is the Jacobian matrix. The control input vector  $\tau$  includes the human joint torque  $\tau_h$  and the torque generated by the exoskeleton  $\tau_e$ , such that

$$\tau = \tau_h + \tau_e. \quad (2)$$

In this paper, a knee joint exoskeleton is used to assist the knee joint flexion/extension movements during sitting position and daily living activities, such as walking and sit-to-stand. In this case, the knee joint dynamics could be rewritten from model (1) as follows:

$$J\ddot{\theta} + B(\dot{\theta} - \dot{\theta}_t) + A\text{sgn}(\dot{\theta} - \dot{\theta}_t) + \tau_g \sin \theta + \tau_l = \tau_e + \tau_h \quad (3)$$

where  $\theta$  and  $\theta_t$  are the knee joint angle and the inclination angle of the reference limb (thigh/shank) in the sagittal plane, respectively,  $J = J_h + J_e$  is the inertia of the human limb/exoskeleton,  $A$  is the solid friction parameter of the exoskeleton,  $B = B_h + B_e$  is the viscous friction parameters of the human knee joint/exoskeleton,  $\tau_g = \tau_{g,h} + \tau_{g,e}$  is the gravitational torque of the human limb/exoskeleton,  $\tau_e$  and  $\tau_h$  are the torques generated by the exoskeleton and by the human muscles, respectively,  $\text{sgn}(\cdot)$  is the sign function, and  $\tau_l$  is the lumped torque caused by the other body segments and the extra forces. For simplification,  $\tau_l$  is considered as a torque that can be estimated using model-based or sensor-based methods [1], [8], [19].

Note that the movements of the exoskeleton and the wearer's shank are considered to be synchronous and simultaneous (i.e., the human limb and the exoskeleton are rigidly coupled). Despite its simplicity, the model (3) could denote the impedance of the knee joints during walking and is enough to explain the IRC issues.

### III. IMPEDANCE REDUCTION CONTROL STRUCTURE

To effectively reduce the human effort needed to move the limb toward a desired movement, one solution is to make the exoskeleton to compensate not only its own impedance but also the partial impedance of the human limb. The goal of this paper is to control the exoskeleton in such a way as to reduce the impedance of the human-exoskeleton system to a desired



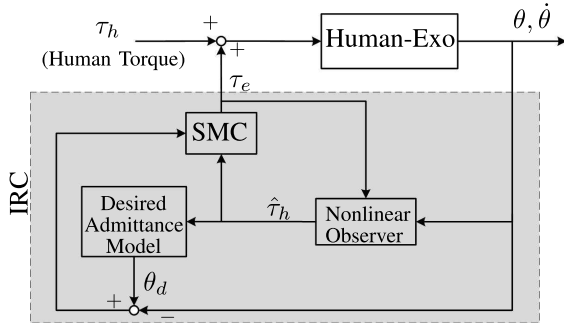


Fig. 3. Nonlinear IRC structure (SMC: sliding-mode-based controller, desired admittance model:  $1/Z_d$ .  $\hat{\tau}_h$ : estimated human torque, and  $\theta_d$ : desired joint angle).

level that is lower than the original impedance of the human limb (see Fig. 2).

Fig. 3 shows the proposed nonlinear IRC structure, which consists of three parts: a nonlinear human torque observer, a desired admittance model (i.e., the inverse of a desired impedance model  $1/Z_d$ ), and an SMC. The principle of this structure is as follows: the nonlinear observer is used to estimate the human joint torque ( $\tau_h$ ), and the SMC is used to track the reference motion trajectory ( $\theta_d$ ) generated by the desired admittance model. The original human-exoskeleton model can be adapted to the desired admittance model with respect to the estimated human torque ( $\hat{\tau}_h$ ) if the following conditions are satisfied: 1) the nonlinear observer can accurately estimate the human joint torque ( $\hat{\tau}_h = \tau_h$ ) and 2) the SMC shows a high tracking accuracy ( $\theta = \theta_d$ ). In Sections III-A–III-C, each part of the impedance control structure is developed. The performance and robustness with respect to modeling uncertainties are analyzed as well.

#### A. Human Torque Observer

Let  $x = [\theta, \dot{\theta}]^T$  be the state vector of the human-exoskeleton system. The dynamic model (3) can be rewritten as follows:

$$\dot{x} = F(x) + G(x)(\tau_e + \tau_h) \quad (4)$$

where

$$F(x) = \begin{bmatrix} \dot{\theta} \\ \frac{1}{J}(-\tau_g \sin \theta - A \operatorname{sgn}(\dot{\theta} - \dot{\theta}_t) - B(\dot{\theta} - \dot{\theta}_t) - \tau_l) \end{bmatrix}$$

$$G(x) = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix}.$$

From the human-exoskeleton system's viewpoint, the human joint torque  $\tau_h$  can be considered as an external disturbance. It can be estimated using the NDO technique [25]–[27]. According to our previous study [23], a nonlinear human-joint-torque observer is designed as follows:

$$\begin{cases} \hat{\tau}_h = z + p(x) \\ \dot{z} = L(-F(x) - G(x)(\tau_e + z + p(x))) \end{cases} \quad (5)$$

where  $\hat{\tau}_h$  is the estimation of the human input torque  $\tau_h$ .  $p(x)$  is chosen as  $p(x) = k_1\theta + k_2\dot{\theta}$ , where  $k_1$  and  $k_2$  are two positive constants.  $z$  is an auxiliary variable that is used to avoid the acceleration measurement.  $L$  satisfies

$$L = \frac{\partial p(x)}{\partial x} = [k_1 \ k_2]. \quad (6)$$

From (4) and (5), the dynamics of the estimated human torque are calculated as follows:

$$\dot{\hat{\tau}}_h = LG(\tau_h - \hat{\tau}_h). \quad (7)$$

Define the estimation error of the human torque as  $\tilde{\tau}_h = \tau_h - \hat{\tau}_h$ .

In the design of a conventional NDO, disturbances (e.g., frictions) are usually considered to be constant or to be changing with a small velocity [25]. However, the human torques generated while performing daily living activities generally show relatively high variability  $\dot{\tau}_h$  that cannot be neglected. Assuming that the velocity of the human torque is bounded  $\|\dot{\tau}_h\|_\infty < \dot{\tau}_{h,\max}$ , the following proposition addresses the stability and tracking accuracy of the nonlinear observer. The proof is given in Appendix A.

**Proposition 1:** Assume that the human-exoskeleton system is subject to human torque  $\tau_h$  described by (4), and the change rate of the human joint torque is bounded, i.e.,  $\|\dot{\tau}_h\|_\infty < \dot{\tau}_{h,\max}$ . The nonlinear observer is given in (5). The estimation error  $\tilde{\tau}_h$  is globally uniformly ultimately bounded  $\|\tilde{\tau}_h\| \leq (J\dot{\tau}_{h,\max}/k_2)$ .

Note that the estimation error is determined by the human torque velocity and the parameter  $k_2$ . Although the estimation error can be restricted to a relatively small range by choosing an appropriate high value of the parameter  $k_2$ ; however, the resulting measurement noise of the angular velocity  $\dot{\theta}$  is added to the estimated torque  $\hat{\tau}_h$ . Hence, a tradeoff between the torque estimation accuracy and the noise suppression should be made for the purpose of practical experimentation. In this paper, the value of the parameter  $k_2$  is chosen based on the main frequency range (0.5–1.4 Hz, [15]) of the human lower limb.

#### B. Sliding-Mode-Based Controller

An NDO-based SMC is used to ensure a high control gain. From the exoskeleton's point of view, the human effort is considered as an external disturbance in the control scheme. Hence, the human joint torque observer can also be used as a disturbance observer in the exoskeleton controller. To design the NDO-based SMC, the first-order sliding surface is used as follows:

$$s = c_1 e + \dot{e} \quad \text{with } e = \theta - \theta_d \quad (8)$$

where  $\theta_d$  denotes the desired trajectory and  $c_1$  is a positive constant.

The NDO-based SMC is then designed as follows:

$$\tau_e = -Jc_1\dot{e} + \tau_g \sin \theta + A \operatorname{sgn}(\dot{\theta} - \dot{\theta}_t) + B(\dot{\theta} - \dot{\theta}_t) + \tau_l - \hat{\tau}_h + J\ddot{\theta}_d - \lambda s - \sigma \operatorname{sgn}(s) \quad (9)$$

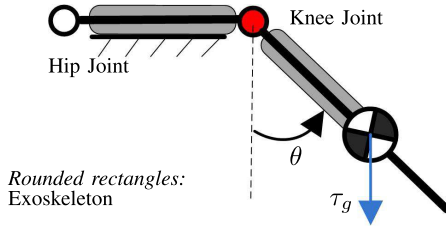


Fig. 4. Schematic of a subject wearing the EICOSI exoskeleton in the sitting position.  $\theta$ : joint angle and  $\tau_g$ : gravitational torque of the human shank and foot and exoskeleton.

with

$$\lambda = \lambda_0 \frac{1 + \beta_1 |e|}{1 + \beta_2 |\dot{\hat{t}}_h|}, \quad \sigma = \begin{cases} \sigma_0, & |e| \geq E_0 \\ 0, & |e| < E_0 \end{cases} \quad (10)$$

where  $\lambda_0, \beta_1, \beta_2$ , and  $E_0$  are positive constants.  $\sigma_0 > |\tilde{\tau}_{h,\max}|$ .  $|\tilde{\tau}_{h,\max}|$  represents the upper bound of the estimation error of the torque observer.

$\lambda$  increases with the tracking error  $e$  and decreases with the derivation of the estimated human torque  $\hat{t}_h$ . Additionally, the nonlinear control term  $\sigma \text{sgn}(s)$  is used only when the tracking error  $e$  exceeds a given range  $E_0$ . The use of the designed  $\lambda$  and  $\sigma$  value as shown in (10) can significantly reduce the effects of the SMC chattering on the estimated human torque and limit the tracking error to a small range. The stability of the exoskeleton system using the disturbance-based SMC is analyzed in the following proposition, and its proof can be found in Appendix B.

**Proposition 2:** Consider the exoskeleton system (3) with the DOB-based SMC (9). Assume that the estimation error of the human torque  $\tilde{\tau}_h$  is bounded; then, the tracking error  $s$  is globally uniformly ultimately bounded. When the rate of change of the human torque is negligible (i.e.,  $\dot{\tau}_h \approx 0$ ), the tracking error of the NDO-based SMC converges to zero.

### C. Desired Admittance Model

According to the IRC structure (see Fig. 3), the desired admittance model  $1/Z_d$  expresses the real joint admittance that the wearer should effectively perceive. The human effort needed to achieve knee joint flexion/extension movements can be reduced by choosing an appropriate small desired impedance  $Z_d$ . To, respectively, test the effect of the inertia, damping, and stiffness reduction independently, an impedance model is chosen as follows:

$$\tau_h = J_d \ddot{\theta} + B_d (\dot{\theta} - \dot{\theta}_t) + K_d \theta \quad (11)$$

where  $J_d, B_d$ , and  $K_d$  represent the desired inertia, damping, and stiffness factors, respectively.

Here, the stiffness is used to denote the gravitational torque acting at the knee joint level. The design of the parameter  $K_d$  determines the level of gravity compensation and can be expressed as follows:

$$K_d = \alpha \tau_g \quad (12)$$

where  $\alpha$  is the compensation ratio.

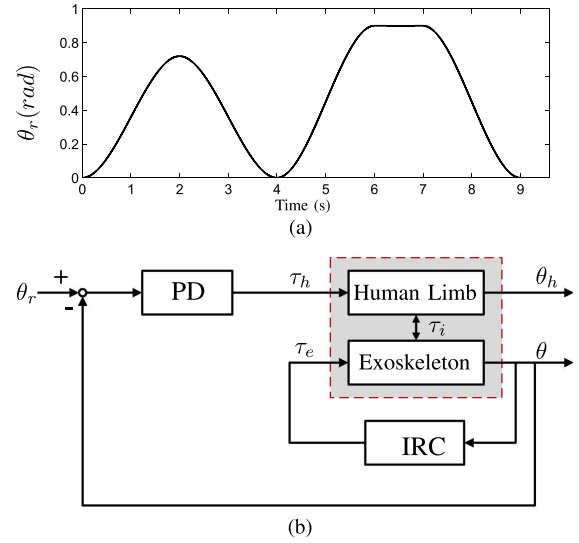


Fig. 5. Control scheme of the simulation. A PD controller is used to simulate the wearer's role for performing a desired movement.  $\tau_i$  shows the interactive torque between the human limb and the exoskeleton. (a) Simulation reference trajectory (0.25 Hz). (b) Control scheme.

### D. Performance of the IRC

To examine the performance of the human joint torque observer and the NDO-based SMC (i.e., the two aforementioned conditions:  $\theta = \theta_d$  and  $\tau_h = \hat{\tau}_h$ ), a sample simulation was carried out. The wearer was considered in a sitting position, i.e.,  $\tau_l = 0$  and  $\dot{\theta}_t = 0$  (see Fig. 4). The human-exoskeleton closed-loop system was controlled using a proportional-derivative (PD) controller to track a given reference angle trajectory, as shown in Fig. 5. Here, the PD controller is only used to simulate the wearer's role, as shown in Fig. 2, and ensure a given joint output trajectory of the human-exoskeleton system. The parameters of the human shank and the exoskeleton system are set as follows:  $J_h = 0.2587 \text{ kg} \cdot \text{m}^2$ ,  $J_e = 0.08 \text{ kg} \cdot \text{m}^2$ ,  $B_h = 1.5569 \text{ Nm} \cdot \text{s} \cdot \text{rad}^{-1}$ ,  $B_e = 0.408 \text{ Nm} \cdot \text{s} \cdot \text{rad}^{-1}$ ,  $\tau_{g,h} = 9.2305 \text{ Nm}$ ,  $\tau_{g,e} = 0.4150 \text{ Nm}$ , and  $A = 0.187 \text{ Nm}$ . The processes used to identify these parameters will be detailed in Section IV-A. The PD controller parameters are obtained using the fine-tune function of the MATLAB for each tracking simulation.

In the simulation, the human-exoskeleton closed-loop system is controlled using the PD controller to track a given reference angle trajectory  $\theta_r$  with four different frequencies (0.25, 0.5, 1, and 2 Hz), as shown in Fig. 5. The desired impedance model  $Z_d$  is set as half of the human joint impedance (i.e.,  $J_d = 0.5J_h$ ,  $B_d = 0.5B_h$ , and  $K_d = 0.5(\sin \theta / \theta) \tau_{g,h}$ ). The parameters of the human joint torque observer (5) and the NDO-based SMC (9) are set as follows:  $k_1 = 100, k_2 = 100, \lambda_0 = 100, \beta_1 = 10, \beta_2 = 1, \sigma_0 = 1$ , and  $E_0 = 0.1$ . As regards the shank and foot/exoskeleton model [see Fig. 5(b)], if the human limb and the exoskeleton are supposed to be rigidly coupled, the human joint angle  $\theta_h$  and the exoskeleton angle  $\theta$  are equal. However, an inevitable misalignment between the human limb and the exoskeleton occurs in practice. Hence, a given modeling error is added in the shank and foot/exoskeleton system in the simulation model.

TABLE I

VERIFICATION OF THE PERFORMANCE OF THE IRC WITH AND WITHOUT MODELING ERRORS.  $e$ : TRACKING ERROR OF THE NDO-BASED SMC (UNIT: rad) AND  $\tilde{\tau}$ : ESTIMATION ERROR OF THE HUMAN JOINT TORQUE (UNIT: Nm)

| Frequency | without    |                       | with       |                       |
|-----------|------------|-----------------------|------------|-----------------------|
|           | RMS( $e$ ) | RMS( $\tilde{\tau}$ ) | RMS( $e$ ) | RMS( $\tilde{\tau}$ ) |
| 0.25-Hz   | 0.00064    | 0.004                 | 0.0013     | 0.32                  |
| 0.5-Hz    | 0.00062    | 0.013                 | 0.0013     | 0.32                  |
| 1-Hz      | 0.00064    | 0.057                 | 0.0011     | 0.44                  |
| 2-Hz      | 0.00350    | 0.347                 | 0.0038     | 2.25                  |

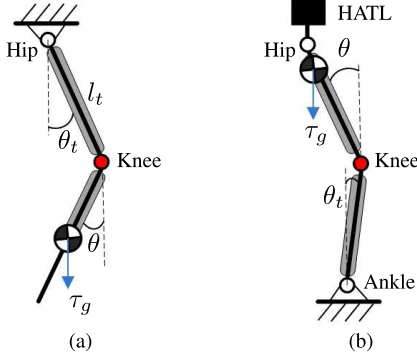


Fig. 6. Schematic view of a subject wearing the EICOSI exoskeleton during two typical walking phases (a) swing and (b) stance phases. HATL indicates the human head, arms, torso, and the swing leg.

A nonlinear relationship between the human joint angle  $\theta_h$  and the exoskeleton angle  $\theta$  is used, i.e.,  $\theta_h = \theta + 0.1 \sin \theta$ , by considering the possible misalignment between the exoskeleton and the human lower limb [28]. Note that the nonlinear term  $0.1 \sin \theta$  has overestimated the misalignment when using the EICOSI exoskeleton. Experimental results show that the RMS of the differences between the exoskeleton angle and the human knee joint angle measured using an electrogoniometer is 0.0384 rad during 10-cycle flexion/extension movements with an amplitude of  $\approx 1.22$  rad in sitting position and 0.0287 rad during 10 walking gait cycles. Table I shows the simulation results including the RMS errors of the human joint torque observer and the NDO-based SMC when tracking the reference as shown in Fig. 5(a), respectively. One can notice that the NDO-based SMC shows both highly accurate performances with and without modeling errors. The estimation error of the human joint torque observer is also accurate without modeling error and is limited to a small range with modeling error. The influence of the modeling error will be detailed and analyzed in Section III-F.

#### E. Application in Walking Activities

When using the proposed IRC to assist the wearer during walking, the human-limb model is no longer invariant, such as the case in the sitting position, and varies with respect to swing phase and stance phase of the gait motion. Correspondingly, the lumped torque  $\tau_l$  should be estimated for each phase, although the model (3) can be used to indicate the knee joint impedance during the stance and swing phases. Here, a model-based method is used to analyze the dynamics of the knee joint

during the two typical walking gait phases (stance and swing phases) (see Fig. 6).

1) *Swing Phase*: During the swing phase, the dynamics of the knee joint is also affected by the movements of the hip joint, and the torque  $\tau_l$  can be expressed as

$$\tau_l = m_s d_s l_t (\ddot{\theta}_t \cos(\theta_t - \theta) - \dot{\theta}_t^2 \sin(\theta_t - \theta)) \quad (13)$$

where  $m_s$  shows the lumped mass of the human shank and the corresponding orthosis segment.  $l_t$  represents the length of the human thigh, while  $d_s$  shows the length of the human shank part from the knee joint center to the center of mass of the shank and foot. Here,  $\theta_t$  is the hip joint angle.

2) *Stance Phase*: During the stance phase, the ankle joint could be considered as the pivot point. Then,  $J$  and  $\tau_g$  in model (3) represent, respectively, the effective inertia and gravitational torque related to the knee joint, i.e.,

$$J = \mathbf{M}(\mathbf{q})_{[2,2]}, \quad \tau_g = \mathbf{G}(\mathbf{q})_{[1,2]} \quad (14)$$

where the subscript  $[i, j]$  represents the element in the row  $i$  and column  $j$ .

Then, the torque  $\tau_l$  can be calculated by

$$\tau_l = \tau_s[1,2] - J\ddot{\theta} - \tau_g \sin \theta \quad (15)$$

with

$$\tau_s = \mathbf{R}(\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) - \mathbf{J}^T \mathbf{F}) \quad (16)$$

where  $\mathbf{R}$  denotes an upper unitriangular matrix.

It should be noted that the torque  $\tau_l$  should be fully compensated by the exoskeleton according to the desired impedance model (11). However, since the torque  $\tau_l$  would be relatively large during the stance phase, it is usually challenged for the exoskeleton actuators to fully compensate it. An alternative method is to include partial estimated torque  $\tau_l$  into the desired impedance model (11). This case has been discussed in a previous work to assist the sit-to-stand movement [22].

Apart from the two models shown in Fig. 6, the human body model also changes with respect to different ground contact conditions. Human body models during walking and other daily living activities, such as stair ascent/descent, have been analyzed in the literature (see [1], [8], [19]). If the human body model is known during a given activity, the torque  $\tau_l$  in model (3) can be consequently estimated.

#### F. Robustness With Respect to Modeling Uncertainties

In general, the torque-sensor-less control structure (i.e., the model-based method) is sensitive to modeling uncertainties. Consider the fact that the human limb parameters are difficult to be estimated accurately (i.e., certain unknown uncertainties are implicitly included in the human-exoskeleton model); therefore, the robustness of the proposed impedance control method with respect to the modeling uncertainties must be guaranteed.

Let us consider that the real-human-exoskeleton system is modeled as follows:

$$J^* \ddot{\theta} + B^* (\dot{\theta} - \dot{\theta}_t) + A^* \text{sgn}(\dot{\theta} - \dot{\theta}_t) + \tau_g^* \sin \theta + \tau_l^* = \tau_e + \tau_h \quad (17)$$

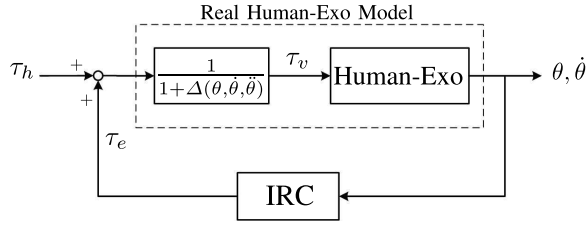


Fig. 7. Human-exoskeleton closed-loop system with modeling uncertainties. Human-Exo denotes the nominal system, i.e., the identified human-exoskeleton system.  $\tau_v$  represents a virtual joint torque.

with

$$\begin{aligned} J^* &= J + \Delta J \\ B^* &= B + \Delta B \\ A^* &= A + \Delta A \\ \tau_g^* &= \tau_g + \Delta \tau_g \\ \tau_l^* &= \tau_l + \Delta \tau_l \end{aligned} \quad (18)$$

where  $J$ ,  $B$ ,  $A$ , and  $\tau_g$  are the identified parameters, and  $\Delta J$ ,  $\Delta B$ ,  $\Delta A$ , and  $\Delta \tau_g$  are their corresponding uncertainties.  $\tau_l$  is the estimated lumped torque caused by the movements of other segments and the external forces, while  $\Delta \tau_l$  is the estimation error.

Hence, the real-human-exoskeleton system (17) can be rewritten as follows:

$$\begin{aligned} J\ddot{\theta} + B(\dot{\theta} - \dot{\theta}_t) + A\text{sgn}(\dot{\theta} - \dot{\theta}_t) + \tau_g \sin \theta + \tau_l \\ = \tau_e + \tau_h + T(\theta, \dot{\theta}, \ddot{\theta}) \end{aligned} \quad (19)$$

where  $T(\theta, \dot{\theta}, \ddot{\theta})$  denotes the lumped modeling uncertainty that is expressed as follows:

$$\begin{aligned} T(\theta, \dot{\theta}, \ddot{\theta}) &= -\Delta J\ddot{\theta} - \Delta B(\dot{\theta} - \dot{\theta}_t) - \Delta A\text{sgn}(\dot{\theta} - \dot{\theta}_t) \\ &\quad - \Delta \tau_g \sin \theta - \Delta \tau_l. \end{aligned} \quad (20)$$

Assume that the modeling uncertainty of each identified parameter is bounded (i.e.,  $\Delta J$ ,  $\Delta B$ ,  $\Delta A$ ,  $\Delta \tau_g$ , and  $\Delta \tau_l$  are all bounded), and the lumped modeling uncertainty is time-varying but with bounded variation, i.e.,

$$\|\dot{T}(\theta, \dot{\theta}, \ddot{\theta})\| < \dot{T}_{\max} \quad (21)$$

where  $\dot{T}_{\max}$  is a positive constant.

It can be observed that there exists  $\Delta(\theta, \dot{\theta}, \ddot{\theta})$ , such that

$$\begin{aligned} \Delta(\theta, \dot{\theta}, \ddot{\theta}) \\ = \frac{\Delta J\ddot{\theta} + \Delta B(\dot{\theta} - \dot{\theta}_t) + \Delta A\text{sgn}(\dot{\theta} - \dot{\theta}_t) + \Delta \tau_g \sin \theta + \Delta \tau_l}{J\ddot{\theta} + B(\dot{\theta} - \dot{\theta}_t) + A\text{sgn}(\dot{\theta} - \dot{\theta}_t) + \tau_g \sin \theta + \tau_l} \end{aligned} \quad (22)$$

and

$$\|\Delta(\theta, \dot{\theta}, \ddot{\theta})\| < \gamma \quad (23)$$

where  $\gamma$  is a positive constant.

The human-exoskeleton system (19) can, then, be expressed as

$$\begin{aligned} \tau_e + \tau_h &= (1 + \Delta(\theta, \dot{\theta}, \ddot{\theta}))(J\ddot{\theta} + B(\dot{\theta} - \dot{\theta}_t) \\ &\quad + A\text{sgn}(\dot{\theta} - \dot{\theta}_t) + \tau_g \sin \theta + \tau_l). \end{aligned} \quad (24)$$

Hence, the impedance control system with modeling uncertainties can be represented as in Fig. 7. It should be noted that the lumped modeling uncertainty  $T(\theta, \dot{\theta}, \ddot{\theta})$  is included into the estimated human joint torque  $\hat{\tau}_h$ , since the identified parameters are used in the human joint torque observer (5). In this case, the estimated human joint torque is given by

$$\hat{\tau}_h = \tau_h + T(\theta, \dot{\theta}, \ddot{\theta}) + e_d \quad (25)$$

where  $e_d$  is the estimation error and is bounded  $\|e_d\| < E_d$  based on Proposition 1 and (21).

Let us define the desired impedance model as follows:

$$\begin{aligned} \hat{\tau}_h &= J_d\ddot{\theta}_d + B_d(\dot{\theta}_d - \dot{\theta}_t) + K_d\theta_d \\ &= \Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)(J\ddot{\theta}_d + B(\dot{\theta}_d - \dot{\theta}_t) + \tau_g \sin \theta_d) \end{aligned} \quad (26)$$

where  $\Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)$  denotes a positive impedance reduction ratio that satisfies  $\|\Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)\| \leq 1$ .

Substituting (25) and (26) into the NDO-based SMC (9), the torque  $\tau_e$  can be rewritten as follows:

$$\tau_e = \left( \frac{1}{\Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)} - 1 \right) \hat{\tau}_h + e_s \quad (27)$$

where

$$\begin{aligned} e_s &= -(J\dot{s} + \lambda s) + A\text{sgn}(\dot{\theta} - \dot{\theta}_t) - \sigma \text{sgn}(s) \\ &\quad + B\dot{e} + \tau_g(\sin \theta - \sin \theta_d) + \tau_l \end{aligned} \quad (28)$$

Since the lumped modeling uncertainty  $T(\theta, \dot{\theta}, \ddot{\theta})$  is also treated as part of the estimated human joint torque  $\hat{\tau}_h$ , the tracking error of the NDO-based SMC (9) is still bounded based on Proposition 2. Here,  $\tau_l$  is also considered to be bounded. Furthermore, one can observe that  $e_s$  is also bounded  $\|e_s\| < E_s$ .

Combining (27) and the human joint torque observer (4), the relationship between the output torque of the NDO-based SMC and the virtual torque  $\tau_v = \hat{\tau}_h + \tau_e$  (see Fig. 7) is given by

$$\tau_e = (1 - \Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d))\tau_v + \Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)e_s \quad (29)$$

where  $\|\Gamma(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)e_s\| \leq E_s$ .

According to the small gain theorem [29], the system is stable if

$$\left\| \frac{1 - \Gamma(\theta, \dot{\theta}, \ddot{\theta})}{1 + \Delta(\theta, \dot{\theta}, \ddot{\theta})} \right\|_{\infty} < 1 \quad (30)$$

Note that the impedance reduction ratio  $\Gamma(\theta, \dot{\theta}, \ddot{\theta})$  should be designed by taking into account the system modeling uncertainty  $\Delta(\theta, \dot{\theta}, \ddot{\theta})$ .

Furthermore, from (25) and (26), the real desired impedance model with respect to the human joint torque  $\tau_h$  can be rewritten as follows:

$$\begin{aligned} \tau_h &= (J_d + \Delta J)\ddot{\theta}_d + (B_d + \Delta B)(\dot{\theta}_d - \dot{\theta}_t) + (K_d\theta_d - \Delta \tau_g \sin \theta) \\ &\quad + \Delta A\text{sgn}(\dot{\theta}_d - \dot{\theta}_t) + \Delta \tau_l - e_d. \end{aligned} \quad (31)$$

Based on Propositions 1 and 2, the estimation error of the human joint observer and the tracking error of the NDO-based SMC are both globally bounded (see Table I). To simplify the analysis, they can be assumed to be equal to zero. According to the discussion at the beginning of Section III,



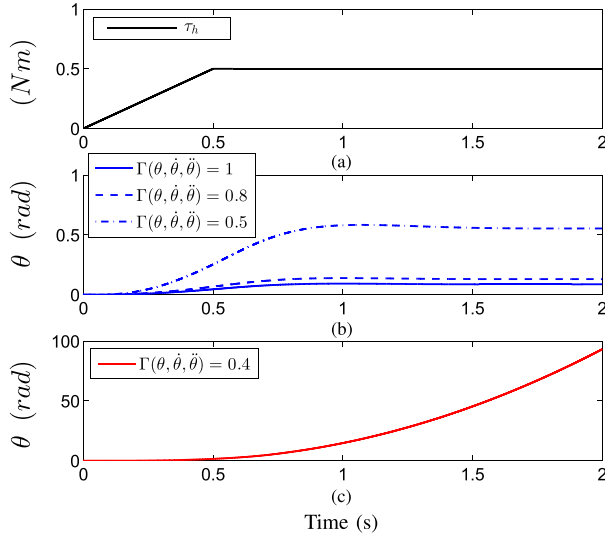


Fig. 8. Evaluation of the robust limitations. A given modeling error ( $\Delta J = -0.4 J$ ,  $\Delta B = -0.4 B$ , and  $\Delta \tau_g = -0.4 \tau_g$ ) is added to the nominal system model.  $\Gamma(\theta, \dot{\theta}, \ddot{\theta})$  denotes the impedance reduction ratios.

the impedance (31) is the real impedance that the wearer perceives during the knee joint flexion/extension movement. One can notice that the real impedance could be affected by the modeling error caused by parametric identification errors and/or knee joint misalignment in exoskeletons. Hence, to ensure a desired real impedance that the wearer perceives, the desired impedance model in IRC can be further tuned based on the identified modeling errors. Furthermore, the condition (30) provides a way to estimate the stability margin of the desired impedance model when the lumped modeling uncertainty  $T(\theta, \dot{\theta}, \ddot{\theta})$  can be estimated.

In particular, in the sitting position (i.e.,  $\tau_l = 0$  and  $\dot{\theta}_l = 0$ ), the stability condition of the human-exoskeleton closed-loop system (see Fig. 7) can also be obtained by analyzing the system passivity with respect to the input human joint torque  $\tau_h$ . If we set  $K_d \theta_d = \alpha \tau_g \sin \theta_d$ , a stability condition is shown as follows:

$$J_d + \Delta J > 0, \quad B_d + \Delta B > 0, \quad \Delta A > 0, \quad \alpha \tau_g + \Delta \tau_g > 0. \quad (32)$$

The proof is given in Appendix C. Note that the stability condition (32) is consistent with (30) obtained using small gain theorem.

Fig. 8 shows the simulation results that illustrate the robustness of the system with respect to the modeling uncertainties. A given modeling error ( $\Delta J = -0.4 J$ ,  $\Delta B = -0.4 B$ , and  $\Delta \tau_g = -0.4 \tau_g$ ) is added to the nominal system model of flexion/extension movements in the sitting position (i.e.,  $\tau_l = 0$  and  $\dot{\theta}_l = 0$ ). According to the human-exoskeleton closed-loop structure shown in Fig. 7, a bounded increasing human joint torque [see Fig. 8(a)] is given with four different impedance reduction ratios ( $\Gamma(\theta, \dot{\theta}, \ddot{\theta}) = 1, 0.8, 0.5, 0.4$ ). It can be observed that when  $\Gamma(\theta, \dot{\theta}, \ddot{\theta})$  does not satisfy the stability condition [see Fig. 8(b)], the human-exoskeleton closed-loop system becomes unstable [see Fig. 8(c)].

TABLE II  
SUBJECTS' INFORMATION

| Subjects | Sex  | Age | Height (cm) | Weight (kg) |
|----------|------|-----|-------------|-------------|
| $S_1$    | Male | 26  | 180         | 78          |
| $S_2$    | Male | 29  | 170         | 66          |
| $S_3$    | Male | 29  | 180         | 63          |
| $S_4$    | Male | 28  | 168         | 68          |

#### IV. EXPERIMENTAL EVALUATION IN FLEXION/EXTENSION MOVEMENTS DURING SITTING

The efficiency of the proposed IRC is first evaluated in the knee-joint flexion/extension movements during the sitting position. Note that the knee joint movements of the swinging leg during a given gait cycle have a pendulum oscillatory flexion/extension behavior that is close to the natural walking frequency [30]. A schematic of a subject wearing the EICOSI with the shank freely moving around the knee joint is shown in Fig. 4.

The experiments are conducted with four healthy subjects (see Table II). The subjects are able to execute complete flexion and extension movements of the knee joint with neither spasticity nor contracture. All subjects were informed of the experimental protocols and gave their consent before participating in the experiments. Constraints on the knee joint position, velocity, and acceleration were imposed, and the security of the range of motion between full extension ( $90^\circ$ ) and full flexion ( $-30^\circ$ ) was ensured mechanically using mechanical joint limiters. All precautions were taken to not adversely affect the health of the participants who served as research subjects.

##### A. Parametric Identification

The parameters of the human-exoskeleton system (3) include two parts: the exoskeleton and the human limb parameters. To obtain the original impedance model of the human limb, these two parts are, respectively, identified using the following steps.

1) *Exoskeleton Parameters*: During this step, only the exoskeleton is controlled in the sitting position (i.e., the human is not coupled with the exoskeleton), as shown in Fig. 4. Sinusoidal angle trajectories  $\theta_r$  with four different frequencies (0.2, 0.4, 0.6, and 0.8 Hz) are chosen as reference trajectories. The exoskeleton is controlled using a basic proportional controller  $\tau_e = K(\theta_r - \theta)$  to track the reference trajectories. The real joint angle  $\theta$  is recorded. The joint angular velocity and acceleration are calculated by once and twice derivations of the joint angle. According to the system model (3), it can be observed that the variables  $\theta, \dot{\theta}, \ddot{\theta}$ , and  $\tau_e$  are known, whereas the parameters  $J_e, B_e, A$ , and  $\tau_{g,e}$  should be identified. On the basis of the system model (3) and the known variables, a linear least squares optimization algorithm is used to identify the exoskeleton parameters [3]. The identified parameters are shown in Table III.

2) *Human Limb Parameters*: The human limb parameters ( $J_h, B_h$ , and  $\tau_{g,h}$ ) can be obtained using the dynamic model (3) if both the human-exoskeleton parameters and



TABLE III  
SYSTEM'S IDENTIFIED PARAMETERS

| Systems       | Initial ( $Kg \cdot m^2$ ) | Viscous friction ( $Nm \cdot s \cdot rad^{-1}$ ) | Solid friction ( $Nm$ ) | Gravitational torque ( $Nm$ ) |
|---------------|----------------------------|--------------------------------------------------|-------------------------|-------------------------------|
| EICOSI        | 0.08                       | 0.408                                            | 0.187                   | 0.4150                        |
| Subject $S_1$ | 0.2800                     | 1.9119                                           | 0                       | 11.5534                       |
| Subject $S_2$ | 0.2535                     | 1.9599                                           | 0                       | 9.5376                        |
| Subject $S_3$ | 0.3030                     | 2.1350                                           | 0                       | 10.860                        |
| Subject $S_4$ | 0.2260                     | 1.9544                                           | 0                       | 9.3176                        |

the exoskeleton parameters are known. Since the exoskeleton parameters can be obtained using the aforementioned method (see Section IV-A1), only the human-exoskeleton parameters need to be identified. Similarly, the exoskeleton worn by the subjects is controlled to track a sinusoidal knee joint trajectory with four different frequencies (0.2, 0.4, 0.6, and 0.8 Hz). The subjects are asked not to exert any muscular effort (i.e.,  $\tau_h = 0$ ) during the identification process. The joint angles are also recorded during this period. The unknown human-exoskeleton parameters in the model (3) can be identified using the linear least squares optimization algorithm as well. The calculated human limb parameters are shown in Table III. Note that the identified shank inertia  $J_h$  and the gravity torque  $\tau_{g,h}$  are similar to those estimated using the regression anthropometry methods presented in [31]. For example, the identified values of  $J_h$  and  $\tau_{g,h}$  of subject 2 are  $2.2587 \text{ kg} \cdot \text{m}^2$  and  $9.2305 \text{ Nm}$ , respectively, whereas the values estimated using the anthropometry method are  $2.2522 \text{ kg} \cdot \text{m}^2$  and  $9.33 \text{ Nm}$ , respectively.

### B. Verification of the Accuracy of the Human Joint Torque Observer

As mentioned in Section III, the conditions ( $\hat{\tau}_h = \tau_h$  and  $\theta = \theta_d$ ) should be satisfied to ensure an accurate desired impedance. The NDO-based SMC was evaluated in the previous study [24]; the experimental results show that the maximum tracking error can be limited to a given range  $E_0 = 0.2 \text{ rad}$  [see (10)]. In [24], the human joint torque observer was just qualitatively evaluated using EMG signals. According to the simulation in Section III-D, the accuracy of the human joint torque observer is affected by the modeling error and/or misalignment between the human joint angle and the exoskeleton angle. To verify the accuracy of the human joint torque observer, the true human joint torque should be measured. Since the human joint torque is the one directly delivered by human muscles, an available approach is to use the musculoskeletal model-based method. However, this approach is usually very complex in practice since the EMG signals from all associated muscles are generally needed, and the method is sensitive to many factors, such as the sensor positions, calibrations, and parameter identification. Note that the interaction torque between the human leg and the exoskeleton cannot be used to indicate the human joint torque since a part of the human joint torque is consumed to swing the human shank and foot. In this case, an alternative method is used. The exoskeleton worn by the subjects is controlled to track a sinusoidal reference angle trajectory with

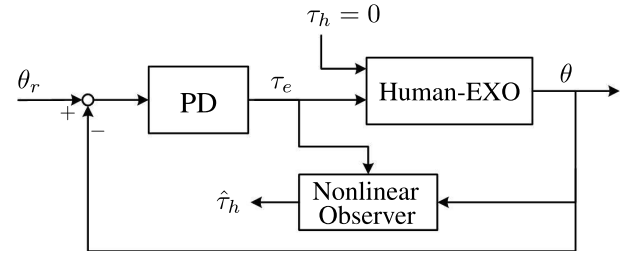


Fig. 9. Control structure for the accuracy verification of the human joint torque observer.

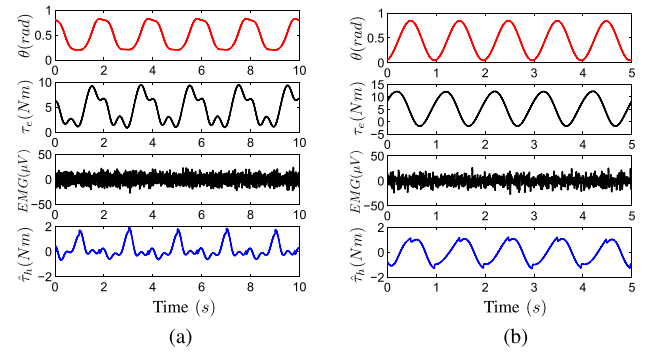


Fig. 10. One subject's experimental results of the accuracy verification of the human joint torque observer. EMG shows the raw EMG signals measured from the VM. (a) 0.5 Hz. (b) 1 Hz.

two different frequencies 0.5 and 1 Hz (see Fig. 9). At the same time, the subjects are asked to not apply any torques, i.e.,  $\tau_h = 0$ . The subjects' effort is monitored using the EMG signals measured from the *vastus medialis* (VM) muscles. Then, the human joint torque is estimated using the proposed observer (5). Fig. 10 shows the experimental results measured with one subject. The average RMS of the estimation error, cross four subjects, is  $\tilde{\tau}_h = 0.99 \pm 0.29 \text{ Nm}$ .

Moreover, the maximum impedance reduction ratio  $\Gamma(\theta, \dot{\theta}, \ddot{\theta})$  should also be determined by the modeling error based on the analysis in Section III-F. To estimate the model errors, a cross validation is also studied. Fig. 11 shows a satisfactory convergence of the measured output torque of the actuator  $\tau_e$  and the estimated output torque  $\hat{\tau}_e$ , obtained using the human-exoskeleton system (3) with the identified parameters. According to (22), the average and maximum magnitudes of the lumped model errors ( $\|\Delta\|_{\text{mean}}$  and  $\|\Delta\|_{\text{max}}$ ) of each subject are calculated, as shown in Table IV. All mean model errors are less than 12%. Based on the stability condition (30), the maximum impedance reduction ratio  $\Gamma(\theta, \dot{\theta}, \ddot{\theta})$  can also be obtained for each subject.

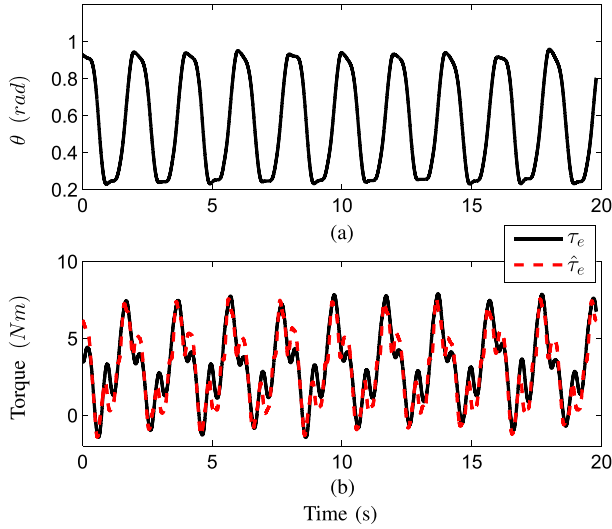


Fig. 11. Cross-validation results of one subject's identified parameters.

TABLE IV  
MAGNITUDES OF THE MODEL UNCERTAINTIES

| Subject | Frequency (Hz) | $\ \Delta\ _{mean}$ | $\ \Delta\ _{max}$ |
|---------|----------------|---------------------|--------------------|
| $S_1$   | 0.5            | 0.12                | 0.47               |
| $S_2$   | 0.5            | 0.10                | 0.45               |
| $S_3$   | 0.5            | 0.08                | 0.23               |
| $S_4$   | 0.5            | 0.08                | 0.28               |

### C. Experimental Protocol

To show the effectiveness of the proposed IRC structure, two experiments were designed: the first one is used to evaluate the reduction in the human effort with decreasing joint impedance, and the second one is conducted to verify the IRC's ability to separately modulate the desired inertia, damping, and stiffness of the human joint impedance. EMG activity recordings of the muscle groups actuating the knee joint [VM and *biceps femoris* (BF)] are used to evaluate the effectiveness of the human effort reduction. The VM is one of the main muscles that ensure the extension of the knee joint in the sitting position, whereas the BF represents the main muscles involved in the knee joint flexion. EMG signal recordings from these two muscles could give a reliable image of the human generated torque or human effort. Two EMG sensors (Delsys, USA) are used to measure the muscular activities in the following experiments. The measured raw EMG signals are processed using the following steps: high-pass filtering at 30 Hz using the fourth-order Butterworth filter, elimination of the measurement noise, and full-wave rectification.

In the first experiment, four different impedance states were used as follows.

- 1) *Human Joint Impedance State* (i.e.,  $Z_h$ ): The subjects swing their shanks without wearing the exoskeleton. This state is used as a baseline state.
- 2) *Human-Exoskeleton Impedance State* (i.e.,  $Z_h + Z_e$ ): The subjects swing their shanks while wearing the exoskeleton without any power assistance. In this state, the exoskeleton is completely passive.

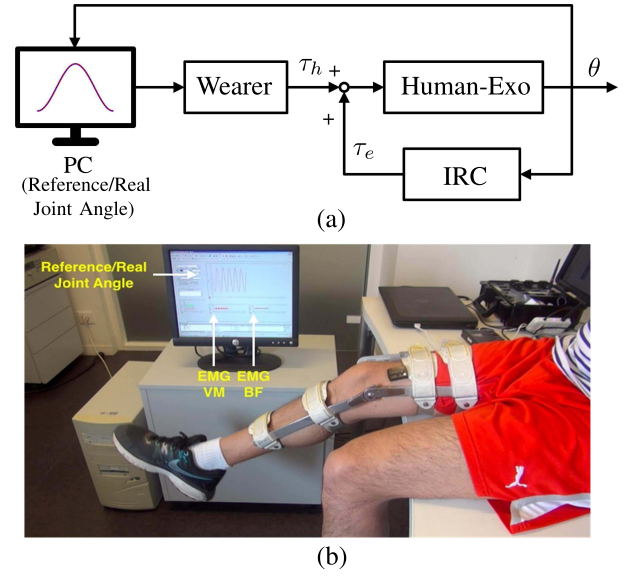


Fig. 12. (a) Control scheme and (b) experimental scenario of the knee joint flexion/extension movements in the sitting position.

- 3) *Exoskeleton Impedance Compensation State* (i.e.,  $Z_d = Z_h$ ): The inertia, friction (damping), and gravity of the exoskeleton are all compensated (i.e., the desired impedance model is equal to the human limb model). In this state, the exoskeleton is considered to be transparent.
- 4) *Human Impedance Compensation State* (i.e.,  $Z_d = 0.5Z_h$ ): The desired impedance of the IRC is set to 50% of the original human joint impedance. Note that the compensation ratio is chosen based on the IRC's robustness with respect to the modeling uncertainties (30) and the real model errors (see Table IV). In this state, the exoskeleton provides an active impedance.

The subjects are asked to move their shanks to track a virtual sinusoidal reference trajectory with 0.5-Hz frequency shown on a host PC display in front of the subject (see Fig. 12). The real knee joint angle is also shown on this PC display. Note that two IMU sensors (Xsens, The Netherlands) attached to the shank and thigh, respectively, are used to measure the knee joint angle during the human joint impedance state. The knee joint angle is calculated based on the IMUs' measured gyros and accelerations [32].

The effectiveness of the human effort reduction is evaluated by comparing the RMS of the EMG signals when the subjects perform the same flexion/extension movement under four different impedance conditions. Consider the fact that the gravity of the human shank and foot is used by the wearer to achieve knee joint flexion movements in the sitting position; the human joint flexion torque is not noticeable when the frequency of the joint movement is low. Hence, the joint extension torque (i.e., the EMG signals from the VM) is used as the main metric, whereas the EMG signals from the BF are measured to monitor the eventual muscles' cocontraction. Correspondingly, the exoskeleton's power assistance is set to zero during flexion movements in all experiments.

In the second experiment, the desired inertia ( $J_d$ ), damping ( $B_d$ ), and stiffness ( $K_d$ ) are further reduced from the

values associated with the exoskeleton impedance compensation state. The effectiveness of the human inertia, damping, and stiffness compensation is evaluated by comparing the RMS of the EMG signals from the VM with those obtained with the exoskeleton impedance compensation state and the human joint impedance state. Furthermore, reference joint angle trajectories with different frequencies are used in this experiment. A low-frequency (0.25 Hz) reference trajectory is used to evaluate the effectiveness of the stiffness compensation. In this case, the influences of the inertia and damping factors are relatively small. Alternatively, a high-frequency (0.8 Hz) reference trajectory is used to show the effectiveness of the inertia and damping compensation.

During all experiments, the EMG signals were recorded, following the successfully tracking of the virtual reference trajectories by the subjects for a certain number of periods ( $>5$  periods). During each of the aforementioned states, two group continuous flexion/extension movements were performed. To avoid the occurrence of muscle fatigue of the subjects during the experiments, each continuous flexion/extension movement lasted no longer than 50 s, and a 1-min interval occurred between the two group movements. At the same time, a 5-min rest interval occurred between the transfer of any two impedance states.

#### D. Experimental Results

Fig. 13 shows the knee joint angles and processed EMG signals of one subject during flexion/extension movements in four states: the human joint impedance state, the human-exoskeleton impedance state, the exoskeleton impedance compensation state, and the human impedance compensation state. Although a certain differences among the knee joint angles  $\theta$  under four states are observed because of the limitation of the wearer's ability for tracking the virtual reference trajectory, the knee joint angles under four states show nearly the same amplitudes and frequencies. At the same time, the measured EMG signals under each impedance state are stable. For all experiments, the RMS of the EMG signals is calculated based on the data measured during 15 flexion/extension movements.

Fig. 14 shows the RMS of the EMG signals of each subject. Table V shows the ratios of the changes in the EMG signals' RMS under the human-exoskeleton impedance ( $Z_h + Z_e$ ), the exoskeleton impedance compensation ( $Z_d = Z_h$ ), and the human joint impedance compensation states ( $Z_d = 0.5Z_h$ ) relative to the values obtained when the subjects were not wearing the exoskeleton ( $Z_h$ ). One can observe that all subjects' EMG signal amplitudes (i.e., joint torques) increased after wearing the exoskeleton ( $Z_h + Z_e$ ), compared with the amplitudes observed without wearing the exoskeleton ( $Z_h$ ). The average increase ratio is  $14.6 \pm 7.3\%$  (see Table V). With the power assistance of the IRC  $Z_d = Z_h$ , the subjects' EMG signals can be reduced to the same or smaller levels than those observed during the human joint impedance state (average increasing by  $-1.8 \pm 4.3\%$ ). Indeed, it can be stated that the human effort required to perform the same knee joint movement is reduced to the level of the state without wearing the exoskeleton. Moreover, all subjects' EMG signals

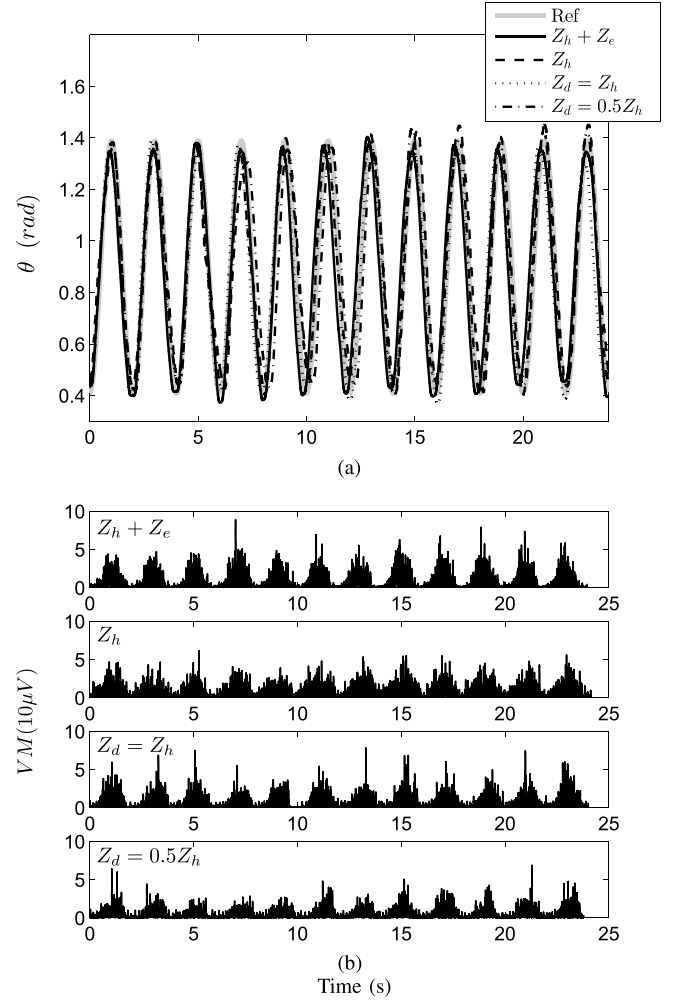


Fig. 13. One group of processed EMG signals during flexion/extension movements in four states. (a) Knee joint angles. (b) Processed EMG signals of the VM.

could be significantly reduced ( $44.7 \pm 2.0\%$ ) when a smaller desired impedance ( $Z_d = 0.5Z_h$ ) is imposed. On the other hand, the average ratio variation of the human joint torques can be predicted using the system model (3) by combining the identified human-exoskeleton parameters. Compared with the human joint torque during the human joint impedance state ( $Z_h$ ), the average ratio variation of human joint torque is as follows:  $Z_h + Z_e$ :  $109.8 \pm 0.008\%$ ,  $Z_d = Z_h$ :  $100\%$ , and  $Z_d = 0.5Z_h$ :  $50\%$ . One can notice that the experiments show consistent results with respect to the predicted ones using the system model (3).

To verify the IRC's ability to modulate the original inertia, damping, and stiffness of the human joint impedance, the human joint impedance state (i.e.,  $Z_h$ ) is chosen as a baseline state as well. During the experiments, the inertia, damping, and stiffness of the impedance model are reduced by 50%, 90%, and 80%, respectively, from the baseline state (i.e.,  $J_d = 0.5J_h$ ,  $B_d = 0.1B_h$ , and  $K_d = 0.2(\sin \theta / \theta) \tau_{g,h}$ ). Note that during each experiment, only one impedance parameter value is reduced; the other two values are kept similar to those of the baseline state. Here, a relatively high reduction ratio (0.5) is used for the inertia reduction since the small

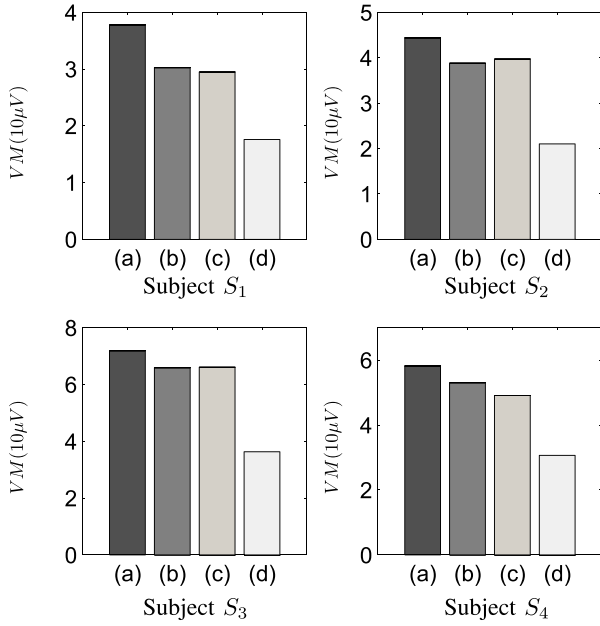


Fig. 14. RMS of the processed EMG signals in four states (a) human-exoskeleton impedance state (i.e.,  $Z_h + Z_e$ ), (b) human joint impedance state (i.e.,  $Z_h$ ), (c) exoskeleton impedance compensation state (i.e.,  $Z_d = Z_h$ ), and (d) human impedance compensation state (i.e.,  $Z_d = 0.5Z_h$ ).

TABLE V  
COMPARISON OF THE RMS OF THE EMG SIGNALS  
MEASURED IN FOUR STATES

| Subjects | States |             |             |                |
|----------|--------|-------------|-------------|----------------|
|          | $Z_h$  | $Z_h + Z_e$ | $Z_d = Z_h$ | $Z_d = 0.5Z_h$ |
| $S_1$    | 100%   | 124.9%      | 97.5%       | 58.1%          |
| $S_2$    | 100%   | 114.5%      | 102.4%      | 54.2%          |
| $S_3$    | 100%   | 109.1%      | 100.2%      | 55.0%          |
| $S_4$    | 100%   | 109.8%      | 92.6%       | 57.8%          |
| Mean     | 100%   | 114.6 %     | 98.2%       | 56.3%          |
| Std      |        | $\pm 7.3\%$ | $\pm 4.3\%$ | $\pm 2.0\%$    |

desired inertia easily causes vibration or instability of the whole human-in-loop control system, as shown in Fig. 2.

The reduction ratios of the RMS of the EMG signals of each subject during stiffness, damping, and inertia compensation are shown in Table VI. Fig. 15 shows the average reduction ratios of the EMG signals' RMS and the ones of the human joint torques calculated using the model (3) during inertia, damping, and stiffness compensation compared with the baseline state. For low-frequency (i.e., 0.25 Hz) joint movements, the EMG signal levels of all subjects are efficiently reduced ( $33.8 \pm 3.1\%$ ) by stiffness compensation [see Fig. 15(a)], whereas the reduction in the EMG signals during damping and inertia compensation is not significant ( $90 \pm 2\%$  and  $89.9 \pm 6.2\%$ ) [see Fig. 15(b) and (c)]. On the other hand, a relatively large reduction in the EMG signal levels ( $69.9 \pm 8.2\%$  and  $75.9 \pm 4.6\%$ ) can be observed during damping and inertia compensation when the subjects swing their legs with a high frequency (0.8 Hz) [see Fig. 15(b) and (c)]. The experimental results fully match the hypothesis considered in the experimental protocol (see Section IV-C). At the same time, the experiments also show consistent results with respect

TABLE VI  
REDUCTION RATIOS OF THE ESTIMATED HUMAN JOINT TORQUES  
AND THE RMS OF THE EMG SIGNALS DURING THE INERTIA,  
DAMPING, AND STIFFNESS COMPENSATIONS

| States                  | Subjects | 0.25-Hz     |             | 0.8-Hz      |             |
|-------------------------|----------|-------------|-------------|-------------|-------------|
|                         |          | Model       | EMG         | Model       | EMG         |
| $0.2\sigma_1\tau_{g,h}$ | $S_1$    | 29.8%       | 30.9%       | —           | —           |
|                         | $S_2$    | 29.2%       | 36.1%       | —           | —           |
|                         | $S_3$    | 30.2%       | 31.4%       | —           | —           |
|                         | $S_4$    | 29.4%       | 37.0%       | —           | —           |
|                         | Mean     | 29.7%       | 33.8%       | —           | —           |
|                         | Std      | $\pm 0.4\%$ | $\pm 3.1\%$ | —           | —           |
| $0.1B$                  | $S_1$    | 91.4%       | 92.0%       | 75.6%       | 78.8%       |
|                         | $S_2$    | 91.6%       | 91.3%       | 75.1%       | 74.0%       |
|                         | $S_3$    | 91.5%       | 88.7%       | 73.2%       | 60.4%       |
|                         | $S_4$    | 90.5%       | 87.9%       | 74.4%       | 66.2%       |
|                         | Mean     | 91.3%       | 90.0%       | 74.6%       | 69.9%       |
|                         | Std      | $\pm 0.5\%$ | $\pm 2.0\%$ | $\pm 1.0\%$ | 8.2%        |
| $0.5J$                  | $S_1$    | 95.2%       | 91.4%       | 85.5%       | 75.8%       |
|                         | $S_2$    | 96.3%       | 89.2%       | 86.3%       | 72.9%       |
|                         | $S_3$    | 94.4%       | 82.1%       | 88.0%       | 73.3%       |
|                         | $S_4$    | 95.9%       | 97.0%       | 86.5%       | 82.6%       |
|                         | Mean     | 95.5%       | 89.9%       | 86.6%       | 75.9%       |
|                         | Std      | $\pm 0.8\%$ | $\pm 6.2\%$ | $\pm 1.0\%$ | $\pm 4.6\%$ |

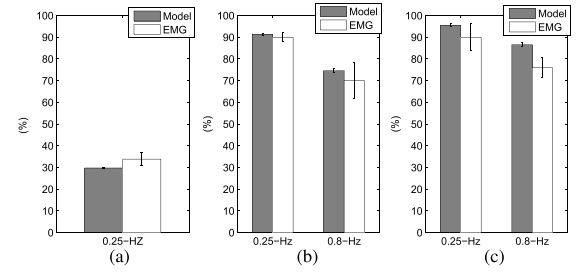


Fig. 15. Reduction ratios of the RMS of the EMG signals during inertia, damping, and stiffness compensation compared with the baseline state  $Z_d (J_h, B_h, \sigma_1\tau_{g,h})$ , where  $\sigma_1 = \sin(\theta)/\theta$ . (a)  $Z_d (J_h, B_h, 0.2\sigma_1\tau_{g,h})$ . (b)  $Z_d (J_h, 0.1B_h, \sigma_1\tau_{g,h})$ . (c)  $Z_d (0.5J_h, B_h, \sigma_1\tau_{g,h})$ .

to the predicted ones using the system model (3), although they are not exactly the same (see Table VI).

To avoid the influence of the order of the above-mentioned experiments, an extra experimental test was implemented. The subjects were asked to perform the knee joint flexion/extension movements in the state “in exoskeleton without assistance” (the most energy-consuming case) by four rounds. During each round, four group continuous flexion/extension movements were performed, and a 1-min interval occurred between each group movements. At the same time, a 5-min rest interval occurred between each round. The EMG signals show that there is no significant changes ( $p > 0.05$ , cross four subjects) among the four rounds using the one-way variance analysis (ANOVA).

During the experiments, the EMG signals of the BF are measured to monitor the cocontraction of the muscles. The experimental results show that there is no noticeable increase in muscle cocontraction during impedance reduction. On the other hand, the EMG signals of the BF show a slight increase during the knee joint extension movements when the desired impedance is smaller than the original human joint impedance.



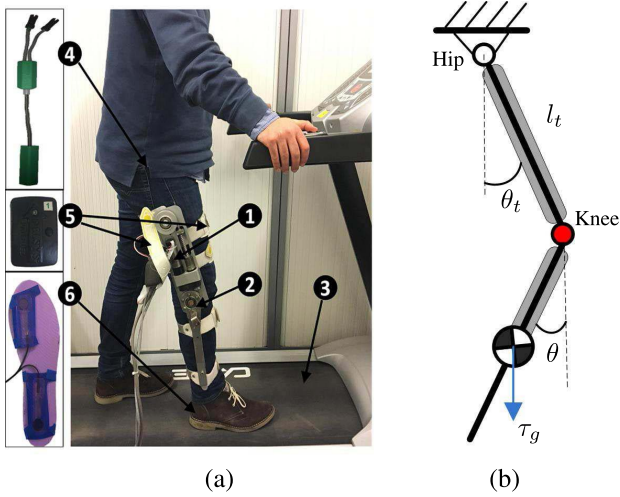


Fig. 16. Subject wearing the EICOSI during walking on a treadmill (a) experimental scenario and (b) knee joint model. 1: knee-joint orthosis. 2: knee joint. 3: treadmill. 4: goniometer. 5: Delsys EMG sensor. 6 insole with two FSR sensors.

At the same time, the EMG signals of the VM are reduced to a smaller level. This means that the subjects produced an additional torque to stop the joint movement when the impedance of the human knee joint is modulated to a smaller level.

## V. EXPERIMENTAL EVALUATION IN SWING MOTION DURING WALKING

Based on the experimental results obtained during the knee-joint flexion/extension movements, the proposed IRC is further evaluated in the swing motion during normal walking (see Fig. 16). As regards the parametric identification of the lumped torque  $\tau_l$  shown in (13), the length of the human thigh  $l_t$  can be measured directly, while the term  $m_s d_s$  can be calculated based on the identified gravity torque  $m_s d_s = \tau_g / g$ . The wearer's hip joint angle is measured using an electrogoniometer. Furthermore, to detect the swing phase during walking, two force-sensitive-resistor (FSR) sensors attached on an insole are used: one FSR sensor is mounted under the toe and the other under the heel.

### A. Experimental Protocol

During the experiments, all subjects are asked to walk on a treadmill with a fixed speed (3 Km/h) (see Fig. 16) under the three states: without wearing the exoskeleton (i.e.,  $Z_h$ ), in exoskeleton without assistance (i.e.,  $Z_h + Z_e$ ), and in exoskeleton with assistance ( $Z_d = 0.7Z_h$ ). At the same time, all subjects are asked to perform the similar gait pattern during different states. EMG signals measured from VM are also used to evaluate the human effort. Under each state, all subjects are asked to walk for 30 s, and a 5-min rest interval is taken between the changes of the states to avoid the influence caused by the muscles' fatigue.

### B. Experimental Results

Fig. 17 shows the ratios of the changes in the EMG signals' RMS, in average, under the human-exoskeleton

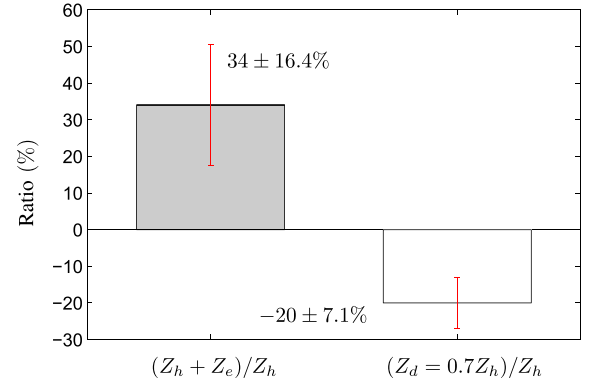


Fig. 17. Reduction ratios of the RMS of the EMG signals during walking compared with the baseline state  $Z_h$  (average ratios cross four subjects).

TABLE VII

PERFORMANCE COMPARISON (MAXIMUM ACTIVE IMPEDANCE COMPENSATION) WITH SENSOR-BASED METHODS

|                            | Gravity | Damping  | Inertia          |
|----------------------------|---------|----------|------------------|
| The IRC                    | 94%     | 100%     | 50%              |
| Ollinger. et. al [14]–[16] | 100%    | 100-140% | $\approx 46\%^*$ |

\* Note that the active inertia reduction was achieved by a model based method, rather than sensor based one which affects the stability because of the sensor non-collocation [15].

impedance ( $Z_h + Z_e$ ), the exoskeleton impedance compensation ( $Z_d = Z_h$ ), and the human joint impedance compensation states ( $Z_d = 0.5Z_h$ ) relative to the values obtained when the subjects were not wearing the exoskeleton ( $Z_h$ ). As mentioned in the experimental protocol, only the EMG signals measured during the swing phase are taken into account. It can be observed that more human efforts are required ( $34 \pm 16.4\%$ ) when wearing a passive exoskeleton comparing to the baseline state. The human effort can be reduced by  $-20 \pm 7.1\%$  when the apparent impedance of the exoskeleton is active (i.e.,  $Z_d = 0.7Z_h$ ).

## VI. DISCUSSION

In [14]–[16], a torque-sensor-based method is used to achieve the same purpose, such as active inertia, damping, and stiffness compensations under the similar experimental conditions, i.e., a knee joint flexion/extension in the sitting position. By comparing to the sensor-based control method, one can also notice that the equivalent effectiveness can be obtained using the proposed IRC, as shown in Table VII. Note that the active inertia reduction cannot be achieved by the sensor-based method in [15] since the sensor noncollocation affects the stability.

From the wearer's point of view, the proposed IRC is also a trajectory-free method. It can also be used to assist the wearer in performing various activities of daily living, such as stance motion of walking, sit-to-stand, and upper-limb movements. For example, in our recent study [8], an extended multijoint IRC structure was used to assist the wearer to perform the sit-to-stand movement using a full lower limb exoskeleton. Unlike the human-exoskeleton model used during the knee

joint flexion/extension sitting position and the swing motion during walking, a tripe inverse pendulum model is used. At the same time, assistances are provided at both the knee and hip joint level. The extra ground reaction force  $\mathbf{F}$  are also taken into account. Regarding the assistance at the knee joint level, the lumped torque  $\tau_l$  in model (3) cannot be fully compensated by the exoskeleton due to the limitation of the output power of the actuator. However, the experimental results show that the human effort can also be reduced compared with the case of wearing the exoskeleton with no assistance.

Regarding the stability and safety of the human-exoskeleton closed-loop system during the experiments, the desired impedance is chosen based on the IRC's robustness with respect to the modeling uncertainties. Based on the stability conditions as analyzed in Section III-F, one can notice that the stability can be guaranteed in two ways. In the first one, more accurate parametric identification methods can be used to reduce the modeling error, in particular, the modeling errors resulting from the human limb parameters identification. For example, Bonnet *et al.* [33] proposed an optimal method to identify the inertial parameters of the body segments, which is more accurate than using the conventional anthropometric tables provided in [31] and [34]. In the other way, the closed-loop system stability can be guaranteed by selecting an appropriate impedance reduction ratio depending on the amplitude of modeling errors. However, it should be noted that the amplitudes of the modeling errors are usually not fixed and change with the frequencies of the human limb movements. Hence, a time-varying impedance reduction ratio can be used to maximize the effectiveness of the impedance reduction without affecting the system stability.

## VII. CONCLUSION

In this paper, a nonlinear IRC structure of the knee-joint exoskeleton is proposed to reduce the human effort by supplying active impedance assistance. The control structure consists of three parts: a nonlinear joint torque observer, an NDO-based sliding mode controller, and a desired admittance model. The proposed strategy adapts the impedance model of the human-exoskeleton system to a desired level that is smaller than the original human limb impedance level. A theoretical analysis of the characteristics and robustness of the system with respect to model uncertainties of the proposed control structure is given. Overall, the main advantages of the proposed control strategy over the existing ones are as follows.

- 1) The hardware requirements of the exoskeleton are relatively small compared with the existing methods; no human-exoskeleton interaction torque is needed to be measured.
- 2) From the application point of view, it is convenient to adjust the inertia, damping, and stiffness of the resulting impedance based on the proposed structure.
- 3) A relatively low impedance can be achieved using the desired impedance model. When the traditional impedance compensation methods based on torque sensors are used, the impedance reduction is usually limited due to the stability problem caused by the sensor non-collocation [15].

The effectiveness of the proposed method was evaluated by assisting four healthy subjects, as they performed flexion/extension movements in the sitting position as well as the swing motion during walking. To assess the level of human torque developed at the knee joint level, the RMS of the processed EMG signals during each shank swing movement was chosen as a metric. In the experiments, while the subjects are in the sitting position, four states were defined to show that the proposed method can mask the impedance of the exoskeleton and reduce the human effort to the same or even lower level than the one required without wearing the exoskeleton. Promising results were obtained during the experiments. With a desired impedance equal to half of the original human joint impedance, the RMS of the EMG signal can be reduced by approximately 44.7%. The ability of the proposed method to separately adjust inertia, damping, and stiffness was also demonstrated through experiments. Furthermore, the experimental results show that the EMG signals can be reduced by  $20 \pm 7.1\%$  in swing motion during walking with the active impedance assistance.

## APPENDIX A

### STABILITY ANALYSIS OF THE HUMAN TORQUE OBSERVER

*Proof:* Choose the Lyapunov function  $W$  as

$$W = \frac{1}{2} \tilde{\tau}_h^2. \quad (33)$$

By the derivation of the Lyapunov function  $W$  and using (7), we obtain

$$\begin{aligned} \dot{W} &= \tilde{\tau}_h \dot{\tilde{\tau}}_h \\ &= \tilde{\tau}_h (\dot{\tau}_h - \dot{\hat{\tau}}_h) \\ &= \tilde{\tau}_h (\dot{\tau}_h - LG \tilde{\tau}_h) \\ &\leq \|\tilde{\tau}_h\| \dot{\tau}_{h,\max} - LG \tilde{\tau}_h^2 \\ &= \|\tilde{\tau}_h\| (\dot{\tau}_{h,\max} - LG \|\tilde{\tau}_h\|) \\ &= \|\tilde{\tau}_h\| \left( \dot{\tau}_{h,\max} - \frac{k_2}{J} \|\tilde{\tau}_h\| \right). \end{aligned} \quad (34)$$

Hence,  $\dot{W} \leq 0, \forall \|\tilde{\tau}_h\| \geq (J \dot{\tau}_{h,\max} / k_2)$ . One can observe that the tracking error ( $\tilde{\tau}_h$ ) is globally uniformly ultimately bounded.

## APPENDIX B

### STABILITY ANALYSIS OF THE SLIDING MODE CONTROLLER

*Proof:* Choose the Lyapunov function  $V$  as

$$V = \frac{1}{2} Js^2 + \frac{1}{2} \tilde{\tau}_h^2. \quad (35)$$

By differentiating (35), and using (3) and (8), we get

$$\begin{aligned} \dot{V} &= Js\dot{s} + \tilde{\tau}_h \dot{\tilde{\tau}}_h = s(-\tau_g \sin \theta - \text{Asgn}(\dot{\theta} - \dot{\theta}_t) - B(\dot{\theta} - \dot{\theta}_t) - \tau_l \\ &\quad + \tau_e + \tau_h + Jc_1 \dot{e} - J\ddot{\theta}_d) + \tilde{\tau}_h \dot{\tilde{\tau}}_h. \end{aligned} \quad (36)$$

By substituting (7) and (9) into the derivative of the Lyapunov function, we obtain

$$\dot{V} = -\lambda s^2 + s(\tilde{\tau}_h - \sigma \text{sgn}(s)) + \tilde{\tau}_h (\dot{\tau}_h - LG \tilde{\tau}_h). \quad (37)$$

When  $|e| \geq E_0$ , then,  $\sigma = \sigma_0 > |\tilde{\tau}_{h,\max}|$ . It can be shown that

$$\dot{V} \leq -\lambda s^2 \leq 0 \quad (38)$$

if

$$\|\tilde{\tau}_h\| \geq \frac{J \dot{\tau}_{h,\max}}{k_2}. \quad (39)$$

When  $|e| < E_0$ , then  $\sigma = 0$ . The derivative of the Lyapunov function  $\dot{V}$  becomes

$$\dot{V} = -\lambda s^2 + s\tilde{\tau}_h + \tilde{\tau}_h(\dot{\tau}_h - L G \tilde{\tau}_h). \quad (40)$$

It follows from (40) that:

$$\dot{V} \leq 0 \quad (41)$$

if

$$\|s\| > \frac{1}{\lambda} \|\tilde{\tau}_h\| \geq \frac{J \dot{\tau}_{h,\max}}{\lambda k_2}, \quad \|\tilde{\tau}_h\| \geq \frac{J \dot{\tau}_{h,\max}}{k_2}. \quad (42)$$

Therefore, the tracking error  $s$  is globally uniformly ultimately bounded. According to Proposition 1,  $|\tilde{\tau}_{h,\max}|$  can be restricted to a small range by choosing a high parameter  $k_2$ . In particular, when the rate of change of the human torque is negligible (i.e.,  $\dot{\tau}_h \approx 0$ ), the estimated error of the human joint torque  $\tilde{\tau}_h$  converges to zero. Correspondingly, the tracking error of the NDO-based SMC converges to zero as well.

#### APPENDIX C PASSIVITY PROOF

*Proof:* To analyze the passivity of the human-exoskeleton closed-loop system, including modeling uncertainties, a storage function is chosen

$$V_s = \frac{1}{2}(J + \Delta J)\dot{\theta}^2 + (\alpha\tau_g + \Delta\tau_g)(1 - \cos\theta) \quad (43)$$

which is a positive function, and its derivative with respect to time is written as follows:

$$\dot{V}_s = (J + \Delta J)\dot{\theta}\ddot{\theta} + (\alpha\tau_g + \Delta\tau_g)\sin(\theta)\dot{\theta}. \quad (44)$$

By substituting (31), we obtain

$$\dot{V}_s = \dot{\theta}\tau_h - (B + \Delta B)\dot{\theta}^2 - \Delta A \operatorname{sgn}(\dot{\theta})\dot{\theta} \leq \dot{\theta}\tau_h \quad (45)$$

that satisfies the condition for the passivity of the system [29].

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